

INTERNATIONAL GCSE **BIOLOGY**

9201/1

Paper 1

Mark Scheme

June 2024

Version: 1.0 Final



2 4 6 Y 9 2 0 1 / 1 / M S

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

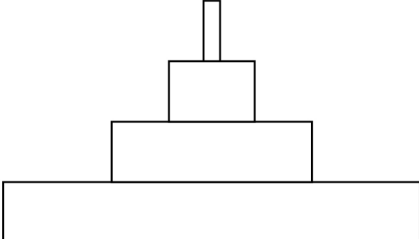
You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	photosynthesis		1	AO1 3.3.1a 3.2.1a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	chlorophyll		1	AO1 3.2.1b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	4-layered triangular pyramid 	allow as blocks or layered triangle, ignore (small) gaps between layers allow inverted pyramid	1	AO2 3.3.1c
	(pyramid) labelled in correct food chain order	all four labels are required	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	10%		1	AO1 3.3.1b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	any one from • in waste materials (from ant) • in movement (of ant) • in respiration (of ant)	ignore heat unqualified allow named waste materials such as faeces / urine / urea / carbon dioxide allow not all (parts) of the ant eaten	1	AO1 3.3.1b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	claws to protect (from predators / meerkat)		1	AO2 3.3.2d
	or to catch / hold prey / ants / food brown body colour to camouflage or to avoid being seen by prey / ants / predator / meerkat	allow blend in with sand / background ignore to hide, unqualified	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7	structural		1	AO3 3.3.2f

Total			9	
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Total			7
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	Cell part <pre> graph LR MP[Mitochondria] --- ME[Structure where energy is released] PL[Plasmids] --- LD[Loops of DNA containing genes] RI[Ribosomes] --- SP[Site of protein production] </pre> do not accept more than one line from a box on the left	Description Loops of DNA containing genes Structure to control what goes into and out of cell Structure where sugar is made from carbon dioxide and water Structure where energy is released Site of protein production	1 1 1	AO1 3.1.1a, c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	spreading out of particles / molecules / ions / atoms from high to low concentration	allow movement of particles / molecules / ions / atoms ignore substances allow down the concentration gradient do not accept along concentration gradient	1 1	AO1 3.1.5a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	concentration of the sugar solution		1	AO4 3.1.5e 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	random errors		1	AO4 3.1.5 6.1 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	cytoplasm has shrunk	allow cytoplasm / membrane has pulled away from the cell wall allow vacuole has got smaller allow cell has plasmolysed	1	AO2 3.1.5d,e
	(because) water moves out of cell		1	
	by osmosis	allow by diffusion of water (through a partially / selectively / semi permeable membrane)	1	
	(because water moves) from a dilute to a more concentrated solution	allow from a high(er) concentration of water to a low(er) concentration of water or allow down the water concentration gradient or allow from a high(er) water potential to low(er) water potential or allow from a low(er) sugar concentration to a high(er) sugar concentration	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	(animal) cell (swells and) bursts	allow cell ruptures	1	AO3 3.1.5d,e 3.1.1 a,b
	(because) there are no cell walls present	MP2 dependent on MP1	1	AO2

Total		13
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	(A) epidermal (tissue)		1	AO1 3.1.4a
	(B) palisade (mesophyll)		1	
	(C) spongy (mesophyll)		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	the movement of (dissolved) sugars	allow named sugars found in plants ignore water / ions / amino acids	1	AO1 3.2.2f
	(in) phloem (tissue)		1	
	from the leaves to the growing regions of the plant	allow from leaves to meristem allow from leaves to shoot / root tip allow from leaves to buds / flowers	1	
	from the leaves to the storage organs		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	add all the numbers (for recorded monthly rainfall)		1	AO2 3.2.2 6.3.7
	divide by 10	allow divide by number of measurements	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	leaves need to absorb carbon dioxide through their stomata for photosynthesis		1	AO2 3.2.2d 3.2.1a
	(but at the same time) leaves lose water by transpiration / evaporation (especially in high temperatures)		1	
	(so) only have leaves when rainfall highest (from December to February) to take in more water (from the soil)		1	
	(so) the water intake and loss are balanced	allow (so) tree would not lose water by transpiration / evaporation faster than it can be replaced by the roots	1	

Total		13
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	(small quantities of) dead / inactive / weakened pathogen / microorganism / virus	allow proteins / antigens (from the virus) allow (pieces of) RNA from the virus	1	AO1 3.4.7d,e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	mumps	allow in any order	1	AO1 3.4.7e
	rubella		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3	any two from - sex - waist-hip ratio - ethnicity - health / fitness levels - not recently vaccinated - not recently had influenza - number of people in each group	allow gender allow BMI allow amount of exercise allow concentration / volume of vaccine	2	AO4 3.4.7e 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	any five from support the conclusion - all 3 concentrations were higher in group 1 compared to group 2. - type A concentration of (antibody) was significantly higher in group 1 compared to group 2 do not support the conclusion - type B/C had (only) a small difference in concentration of antibody between the two groups - number of adults for study was only 274 - we do not know the number in each group - only tested once 3 months after vaccination - only trialled on three strains / types of influenza virus - not many variables controlled - only tested adults over age 65	max four marks if only do not support the conclusion given allow use of relevant figures to explain allow use of relevant figures to explain allow small number of people in trial allow could have tested more often allow there are no error bars allow there are no ranges allow there are no SDs	5	AO3 3.4.7e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	bar drawn, in range of 15–75 arbitrary units	ignore shading / width / spacing	1	AO3 3.4.7e 4–5

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.6	as the virus lives / reproduces inside cells so drugs will damage body cells	allow as the virus lives / reproduces inside cells so drugs will not reach the virus allow (high rate of) mutation (of virus) makes drug less effective	1	AO1 3.4.7g

Total		12
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	(A) gall bladder		1	AO1 3.2.4a
	(B) duodenum		1	

Question	Answers	Mark	AO / Spec. Ref.
06.2	Level 3: Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.	5–6	AO1 3.2.4a, b, c
	Level 2: Relevant points (reasons / causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear.	3–4	
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.	1–2	
	No relevant content	0	

	Indicative content Role of lipase • lipase in an enzyme ○ lipase is produced in pancreas and small intestine ○ enzymes have an active site ○ substrate fits into active site • fat digestion occurs in the small intestine ○ fats broken down into fatty acids and glycerol Role of bile • bile made in liver • bile is stored in the gall bladder • released into the duodenum / small intestine. ○ (so) enzymes / lipase work more rapidly / effectively • bile emulsifies fat ○ (so) increasing surface area of fats for (faster) lipase action • bile neutralises stomach acid • bile makes alkaline conditions in the small intestine ○ (so) makes optimum pH for lipase • incorrect pH would denature the enzyme ○ (so) bonds holding shape of active site are broken ○ (so) shape of active site changes Absorption of food products • fatty acids / glycerol are small soluble molecules • (so) are absorbed (from the small intestine) into the blood ○ by diffusion / active transport ○ (and) transported (around the body) in the plasma / blood For level 3, the roles of lipase and bile and absorption must be explained.		
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	1050 – 420 (= 630)		1	AO2 3.2.4 6.3.3
	$\frac{630}{420} \times 100$		1	
	150 (%)		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	(person 2) consumes more (saturated) fat because their diet contains more oils	allow use of comparative figures allow diet contains more sugar which can be converted to fat	1	3.2.3e 3.2.6b AO2
	(so more likely) build-up of fatty material inside blood vessels / arteries		1	AO2
	(so) coronary arteries blocked / narrowed		1	AO2
	(so) less oxygen / glucose delivered to heart <u>muscles</u> (cells) for (aerobic) respiration		1	AO2
	(so) heart (muscle) <u>cells</u> may die (causing heart attack)		1	AO3

Total		16
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	tongue	allow in any order	1	AO1 3.4.2c
	nose	allow mouth	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	impulse passes from sensory neurone to relay neurone		1	AO1 3.4.1d
	(impulse passes) from relay neurone in CNS to motor neurone	allow (impulse passes) from relay neurone in spinal cord to motor neurone	1	
	(impulse passes) from the motor neurone to the effector / muscle		1	
	impulses crosses synapse as a chemical		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.3	$\frac{59\,500\,000}{50\,000}$		1	AO2 3.5.4a 6.3.6
	1190		1	
	$\frac{1190}{11} \times 4$	allow use of incorrectly calculated number of females	1	
	433	allow 432.72... allow 432 allow an answer calculated from incorrectly calculated number of females	1	
	OR			
	59 500 000 / 11			
	5 409 090.91 x 4 = 21 636 363.6			
	21 636 363.6 / 50 000	allow use of incorrectly calculated number of females		
	432	allow an answer calculated from incorrectly calculated number of females		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.4	parents 6 and 7 have disorder X but child 11 does not	allow two parents who have the disorder have a child who does not have the disorder allow parents 6 and 7 have the disorder so if recessive then all of the offspring would have it	1	AO3 3.5.3g 3.5.4a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.5	female gametes – D and d		1	AO2 3.5.3g 3.5.4a
	male gametes – d and d		1	6.3.13 AO2
	derivation of offspring – Dd Dd dd dd	allow correct derivation from incorrect gametes	1	AO2
	identification Dd with condition and dd without condition		1	AO2
	probability 0.5	allow 50%, 2 in 4, 1:1 do not accept 1:2 probability must match derived genotypes	1	AO3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.6	increased muscle <u>contractions</u>	allow faster muscle <u>contractions</u> allow more rapid muscle <u>contractions</u>	1	AO2 3.2.6f, 3.4.4d
	(which) will require more respiration to release more energy (for muscle contraction)	do not accept energy produced / made / created	1	
	(but) respiration also releases thermal energy (increasing body temperature)	allow respiration also releases heat	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.7	any one from - the proteins block the filter(s) in the kidney (so it fails to work) - too much ammonia / urea is formed (rapidly) which is toxic and damages the kidney		1	AO3 3.4.3c, d

Total		20
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